



GE Medical Systems

gemedical.com

Technical Publication

Direction 2330577-100

Revision 0

GE Medical Systems

Lightspeed Ultra 4 to 8 Slice DAS Upgrade

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LANGUAGE

WARNING

- THIS SERVICE MANUAL IS AVAILABLE IN ENGLISH ONLY.
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- DO NOT ATTEMPT TO SERVICE THE EQUIPMENT UNLESS THIS SERVICE MANUAL HAS BEEN CONSULTED AND IS UNDERSTOOD.
- FAILURE TO HEED THIS WARNING MAY RESULT IN INJURY TO THE SERVICE PROVIDER, OPERATOR OR PATIENT FROM ELECTRIC SHOCK, MECHANICAL OR OTHER HAZARDS.

AVERTISSEMENT

- CE MANUEL DE MAINTENANCE N'EST DISPONIBLE QU'EN ANGLAIS.
- SI LE TECHNICIEN DU CLIENT A BESOIN DE CE MANUEL DANS UNE AUTRE LANGUE QUE L'ANGLAIS, C'EST AU CLIENT QU'IL INCOMBE DE LE FAIRE TRADUIRE.
- NE PAS TENTER D'INTERVENTION SUR LES ÉQUIPEMENTS TANT QUE LE MANUEL SERVICE N'A PAS ÉTÉ CONSULTÉ ET COMPRIS.
- LE NON-RESPECT DE CET AVERTISSEMENT PEUT ENTRAÎNER CHEZ LE TECHNICIEN, L'OPÉRATEUR OU LE PATIENT DES BLESSURES DUES À DES DANGERS ÉLECTRIQUES, MÉCANIQUES OU AUTRES.

WARNUNG

- DIESES KUNDENDIENST-HANDBUCH EXISTIERT NUR IN ENGLISCHER SPRACHE.
- FALLS EIN FREMDER KUNDENDIENST EINE ANDERE SPRACHE BENÖTIGT, IST ES AUFGABE DES KUNDEN FÜR EINE ENTSPRECHENDE ÜBERSETZUNG ZU SORGEN.
- VERSUCHEN SIE NICHT, DAS GERÄT ZU REPARIEREN, BEVOR DIESES KUNDENDIENST-HANDBUCH NICHT ZU RATE GEZOGEN UND VERSTANDEN WURDE.
- WIRD DIESE WARNUNG NICHT BEACHTET, SO KANN ES ZU VERLETZUNGEN DES KUNDENDIENSTTECHNIKERS, DES BEDIENERS ODER DES PATIENTEN DURCH ELEKTRISCHE SCHLÄGE, MECHANISCHE ODER SONSTIGE GEFAHREN KOMMEN.

AVISO

- ESTE MANUAL DE SERVICIO SÓLO EXISTE EN INGLÉS.
- SI ALGÚN PROVEEDOR DE SERVICIOS AJENO A GEMS SOLICITA UN IDIOMA QUE NO SEA EL INGLÉS, ES RESPONSABILIDAD DEL CLIENTE OFRECER UN SERVICIO DE TRADUCCIÓN.
- NO SE DEBERÁ DAR SERVICIO TÉCNICO AL EQUIPO, SIN HABER CONSULTADO Y COMPRENDIDO ESTE MANUAL DE SERVICIO.
- LA NO OBSERVANCIA DEL PRESENTE AVISO PUEDE DAR LUGAR A QUE EL PROVEEDOR DE SERVICIOS, EL OPERADOR O EL PACIENTE SUFRAN LESIONES PROVOCADAS POR CAUSAS ELÉCTRICAS, MECÁNICAS O DE OTRA NATURALEZA.

ATENÇÃO

- ESTE MANUAL DE ASSISTÊNCIA TÉCNICA SÓ SE ENCONTRA DISPONÍVEL EM INGLÊS.
- SE QUALQUER OUTRO SERVIÇO DE ASSISTÊNCIA TÉCNICA, QUE NÃO A GEMS, SOLICITAR ESTES MANUAIS NOUTRO IDIOMA, É DA RESPONSABILIDADE DO CLIENTE FORNECER OS SERVIÇOS DE TRADUÇÃO.
- NÃO TENHA TENTADO REPARAR O EQUIPAMENTO SEM TER CONSULTADO E COMPREENDIDO ESTE MANUAL DE ASSISTÊNCIA TÉCNICA.
- O NÃO CUMPRIMENTO DESTA AVISO PODE POR EM PERIGO A SEGURANÇA DO TÉCNICO, OPERADOR OU PACIENTE DEVIDO A CHOQUES ELÉTRICOS, MECÂNICOS OU OUTROS.

AVVERTENZA

- IL PRESENTE MANUALE DI MANUTENZIONE È DISPONIBILE SOLTANTO IN INGLESE.
- SE UN ADDETTO ALLA MANUTENZIONE ESTERNO ALLA GEMS RICHIEDE IL MANUALE IN UNA LINGUA DIVERSA, IL CLIENTE È TENUTO A PROVVEDERE DIRETTAMENTE ALLA TRADUZIONE.
- SI PROCEDA ALLA MANUTENZIONE DELL'APPARECCHIATURA SOLO DOPO AVER CONSULTATO IL PRESENTE MANUALE ED AVERNE COMPRESO IL CONTENUTO.
- NON TENERE CONTO DELLA PRESENTE AVVERTENZA POTREBBE FAR COMPIERE OPERAZIONI DA CUI DERIVINO LESIONI ALL'ADDETTO ALLA MANUTENZIONE, ALL'UTILIZZATORE ED AL PAZIENTE PER FOLGORAZIONE ELETTRICA, PER URTI MECCANICI OD ALTRI RISCHI.

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このサービスマニュアルには英語版しかありません。

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本维修手册仅存有英文本。

非 GEMS 公司的维修员要求非英文本的维修手册时，客户需自行负责翻译。

未详细阅读和完全了解本手册之前，不得进行维修。

忽略本注意事项会对维修员，操作员或病人造成触电，机械伤害或其他伤害。

DAMAGE IN TRANSPORTATION

All packages should be closely examined at time of delivery. If damage is apparent write "Damage In Shipment" on ALL copies of the freight or express bill BEFORE delivery is accepted or "signed for" by a GE representative or hospital receiving agent. Whether noted or concealed, damage MUST be reported to the carrier immediately upon discovery, or in any event, within 14 days after receipt, and the contents and containers held for inspection by the carrier. A transportation company will not pay a claim for damage if an inspection is not requested within this 14 day period.

Call Traffic and Transportation, Milwaukee, WI (414) 785 5052 or 8*323 5052 immediately after damage is found. At this time be ready to supply name of carrier, delivery date, consignee name, freight or express bill number, item damaged and extent of damage.

Complete instructions regarding claim procedure are found in Section S of the Policy And Procedures Bulletins.

14 July 1993

CERTIFIED ELECTRICAL CONTRACTOR STATEMENT

All electrical Installations that are preliminary to positioning of the equipment at the site prepared for the equipment shall be performed by licensed electrical contractors. In addition, electrical feeds into the Power Distribution Unit shall be performed by licensed electrical contractors. Other connections between pieces of electrical equipment, calibrations and testing shall be performed by qualified GE Medical personnel. The products involved (and the accompanying electrical installations) are highly sophisticated, and special engineering competence is required. In performing all electrical work on these products, GE will use its own specially trained field engineers. All of GE's electrical work on these products will comply with the requirements of the applicable electrical codes.

The purchaser of GE equipment shall only utilize qualified personnel (i.e., GE's field engineers, personnel of third-party service companies with equivalent training, or licensed electricians) to perform electrical servicing on the equipment.

IMPORTANT...X-RAY PROTECTION

X-ray equipment if not properly used may cause injury. Accordingly, the instructions herein contained should be thoroughly read and understood by everyone who will use the equipment before you attempt to place this equipment in operation. The General Electric Company, Medical Systems Group, will be glad to assist and cooperate in placing this equipment in use.

Although this apparatus incorporates a high degree of protection against x-radiation other than the useful beam, no practical design of equipment can provide complete protection. Nor can any practical design compel the operator to take adequate precautions to prevent the possibility of any persons carelessly exposing themselves or others to radiation.

It is important that anyone having anything to do with x-radiation be properly trained and fully acquainted with the recommendations of the National Council on Radiation Protection and Measurements as published in NCRP Reports available from NCRP Publications, 7910 Woodmont Avenue, Room 1016, Bethesda, Maryland 20814, and of the International Commission on Radiation Protection, and take adequate steps to protect against injury.

The equipment is sold with the understanding that the General Electric Company, Medical Systems Group, its agents, and representatives have no responsibility for injury or damage which may result from improper use of the equipment.

Various protective materials and devices are available. It is urged that such materials or devices be used.



LITHIUM BATTERY CAUTIONARY STATEMENTS



CAUTION
Risk of

Danger of explosion if battery is incorrectly replaced. Replace only with the same or equivalent type recommended by the manufacturer. Discard used batteries according to the

ATTENTION
Danger

Il y a danger d'explosion s'il y a remplacement incorrect de la batterie. Remplacer uniquement avec une batterie du même type ou d'un type recommandé par le constructeur. Mettre au rebut les batteries usagées conformément aux instructions du fabricant.

OMISSIONS & ERRORS

Customers, please contact your GE Sales or Service representatives.

GE personnel, please use the GEMS CQA Process to report all omissions, errors, and defects in this publication.

Revision History

Revision	Date	Reason for change
0	06/14/2002	Initial Release of Document

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Chapter 1

Mechanical Upgrade

Section 1.0

Steering Guide

Purpose: To configure the hardware of an existing LightSpeed Ultra/Plus system to support 8 Slice Image generation mode. This includes cleaning the MDAS air plenum, filters, chassis interior and exterior. Inspecting and cleaning the convertor boards if necessary. Adding Convertor Cards to the MDAS and removal of specific Gantry Balance Weights.

Inventory Kit Accumulate Cleaning Materials and Tools	☐ Inventory Upgrade Kit ☐ Gather Tool, ESD and Cleaning Supplies	
Clean Air Plenum and Chassis Interior	☐ Remove Air Plenum, filters and Convertors for Cleaning ☐ Clean Dust and Contaminants from Plenum and Chassis	
Install DAS Convertor Cards	☐ Install Original and New Convertor Cards	
Verify Powers Supply Settings	☐ Verify DAS Power Supplies. Adjust as necessary LTSPD ULTRA/PLUS SERVICE MANUAL - GENERAL Direction 2296434-100, REVISION 4, Chapter 8 Gantry, Section 2.1.4, for specifications. LTSPD ULTRA/PLUS SERVICE MANUAL - GENERAL Direction 2296434-100, REVISION 4, Chapter 7 Detector and DAS, Section 3.4 for adjustment procedure details.	
Remove 4 Slice Balance Weights	☐ Remove 4 Slice Gantry Balance Weights ☐ Perform Service Gantry Balance . Follow software tool instructions	
Configure Software	☐ Perform Reconfig LightSpeed Ultra/Plus Software Installation Procedures, direction 2296436-100, Revision 02 for detailed reconfig instructions. ☐ Perform Flash Download LTSPD ULTRA/PLUS SERVICE MANUAL - GENERAL Direction 2296434-100, REVISION 4, Chapter 2, Section 3.2.2 Flash Download Tool for details. ☐ Install 8 Slice Scanning Option from MOD ☐ Update Scanning Protocols LightSpeed 3.X Application Tips and Work-Arounds, Direction 2296447-100, Rev 3, page 12 for details on updating the scanning protocols.	
Perform Verification Tests	☐ Perform DASTOOLS Interconnect and XRay Ver Tests ☐ Perform DASTOOLS Pop Noise & Microphonics Tests ☐ Correct Any Failures	
Perform Complete Calibrations	☐ Perform DASGAIN CAL ☐ Perform COLLIMATOR CAL ☐ Perform Detailed Phantom Calibrations ☐ Perform FASTCAL ☐ Perform CT Number Adjust	LTSPD ULTRA/PLUS SERVICE MANUAL - GENERAL, DIRECTION 2296434-100, REVISION 4, Chapter 2, Section 2.0 Scanner Utilities for Calibration details.
Perform 1x Image Series and Evaluate image Quality	☐ Perform 1x Image Series ☐ Evaluate Image Quality	LTSPD ULTRA/PLUS SERVICE MANUAL - GENERAL, DIRECTION 2296434-100, REVISION 4, Chapter 11, Image Quality for details.

Section 2.0 Required Parts

UPGRADE TO 2281177 (8 SLICE) -> Part Number 2330483

Part Number	Quantity	Description
2316996		PATIENT 8 SLICE OPTION
2142726	1	DICOM MAGNETO OPTICAL DISK
46-302200P7	1	SERIAL #, MODEL#: RATING PLATE (2316996)
2330482		KIT, MFG 4 TO 8 SLICE MDAS UPGRADE
2258273	24	MDAS CONVERTER CARD
46-302200P7	3	KIT, 4 TO 8 SLICE MDAS UPGRADE rating plate (2330482)
2330577-100		Lightspeed Ultra 4 to 8 Slice DAS Upgrade Manual
2227089-2		DAS/DETECTOR SERVICE KIT
46-183039P1	24	ALCOHOL PADS
2169110-2	1	ONE PKG OF 25 PC ELASTOMER
2207303-6	1	SAFE SKIN NITRILE GLOVE, Large
2207303-7	1	SAFE SKIN NITRILE GLOVE, Extra Large
2185881	8	ESD BOOT-NP-DET-TOOL
2335064	1	DUSTER-AIR SPRAY SYSTEM
2203205	20	ANTISTATIC BAG, 7 X 20"
2254888-2	1	GB TWIN ELAST. (25PCS) IN
46-302200P3		UPGRADE TO 2281177 (8 SLICE) rating plate (2330483)

Section 3.0

Recommended Tools

- 1.) **ESD** Materials (Wrist strap, mat, etc.)2220482
- 2.) Torque Wrench Kit
- 3.) Ratchet Drive Set 3/8"
- 4.) Extension 3/8" x 8"
- 5.) 5 mm hex key for 3/8" socket drive
- 6.) Screw driver (flat) size # 6
- 7.) Vacuum: It is suggested to use the following vacuum:
 - a.) **ESD** Type Vacuum Cleaner (120VAC)46-194427P392
 - b.) **ESD** Type Vacuum Cleaner (220/240VAC)46-194427P391
- 8.) Filters and accessories:
 - a.) Vacuum Cleaner: Anti-Static Dusting Brush, 46-194427P423
 - b.) **ESD** Type Vacuum Cleaner Filter, (10 per box) 46-194427P393
- 9.) Lint free Towels. GE Part Number: 46-307237P1 or equivalent
- 10.) Amax Contact, and Circuit Cleaner, Part #362-16, **MSDS** #:8365595
GE Part Number: 2274104
- 11.) **DAS/DET** Interface Kit: GE Part Number: 2227089-2
Contains Nitrile Gloves, Aero Duster Spray System for use with compressed air and static bags needed by this procedure.

Section 4.0

Procedure Details

4.1 System Preparation

- 1.) Save System State Files. Select All
- 2.) Remove gantry side, top and front covers.
- 3.) Turn off Gantry power at STC. Axial Enable, HVDC Enable, and 120 Vac.
- 4.) Rotate gantry so the MDAS center chassis is at 12'o'clock position and engage Gantry Rotational Lock.



WARNING GANTRY WILL BECOME UN-BALANCED IN THE NEXT STEPS AND WILL ROTATE UNEXPECTEDLY. THIS CAN RESULT IN PERSONAL INJURY. MAKE SURE YOU ENGAGE THE GANTRY ROTATIONAL LOCK.

4.2 DAS Plenum Removal & Cleaning

Comment: A clean environment is important to the success of this procedure. You need to remove as much dust and dirt as possible. Not the following cleaning procedures will result in microphonics failures.

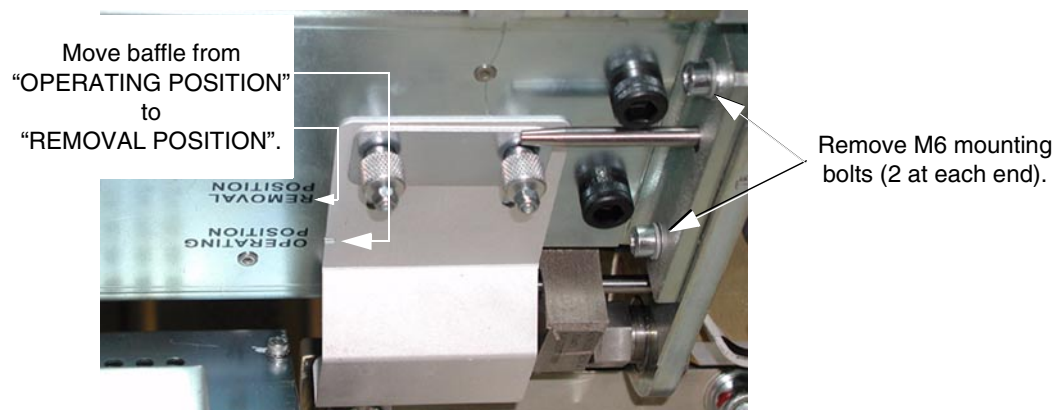


Figure 1-1 DAS Air Plenum: Baffle & Mounting Bolts

- 1.) Disconnect Fan AC power at Connector J22.
- 2.) Set air baffles on either end of the duct to the "Removal Position" (see [Figure 1-1](#)):
 - a.) Loosen knurl nuts
 - b.) Slide baffle into removal position
 - c.) Hand tighten knurl nuts
- 3.) Remove four (4) M6 mounting bolts and lock washers (2 at each end). See [Figure 1-1](#).
- 4.) Carefully slide the plenum towards the front.

Be careful not to catch wires, detector flexes or plenum gaskets.
- 5.) Disconnect MDAS Plenum connections and remove plenum.
- 6.) Remove plenum and set aside on clean surface.
- 7.) Remove filters from plenum and thoroughly clean them.
 - a.) Vacuum filters first to remove as much dust as possible.
 - b.) Wash in warm clean water to remove remaining fine dust particles.

- c.) Place filters aside to dry.
Remove excess water with towels or shaking.
- 8.) Clean MDAS plenum and place aside.
 - a.) Vacuum all surfaces inside and out.
 - b.) Clean fan blades and finger guards.
 - c.) Clean EMI gaskets

4.3 Cleaning the DAS Chassis Interior

4.3.1 ESD Preparation and Converter Card Removal

- 1.) Assemble Anti-Static Kit materials.
 - a.) Anti-Static mat.
 - b.) Anti-Static Converter Card bags.
 - c.) Beeping ground strap monitor.
- 2.) Connect ESD strap and ground to your person and ground point of rotating gantry.
Ensure a good ground is present, no audio alert from ESD strap electronics.
- 3.) Remove the MDAS chassis covers.
Remove dust from covers and gaskets using a vacuum.

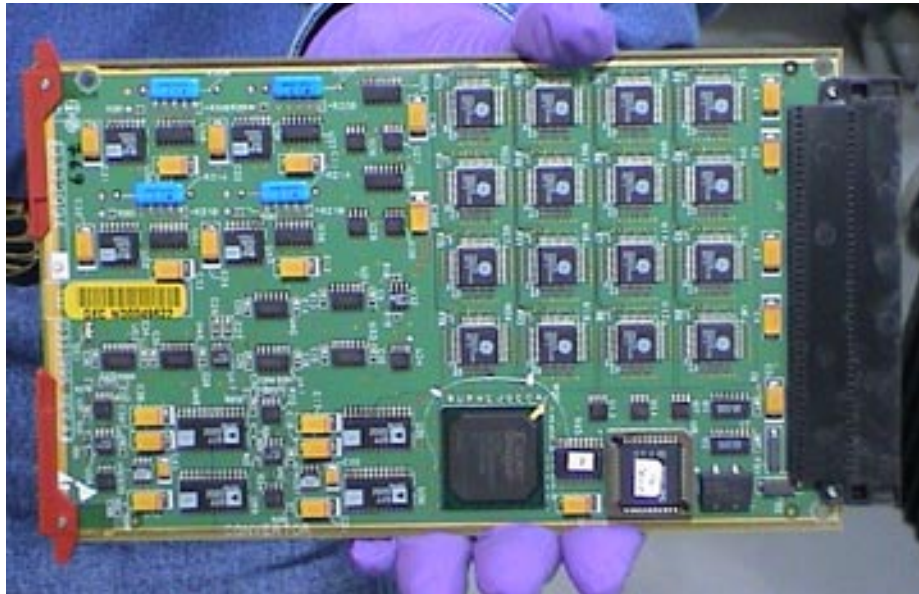


Figure 1-2 Proper method for handling converter cards

- 4.) Remove converter cards 1 at a time and inspect for dust/dirt build up.
 - Handle cards as shown in [Figure 1-2](#).
 - If extreme dust/dirt build up is present, reference .
- a.) Cleaning the Converter Boards:
 - I.) Make sure you are fully grounded using your **ESD** strap.
 - II.) Hold one board at a time, and blow compressed air across the board, removing the dirt/ dust that may have accumulated between component pins, etc. [See Figure 1-3](#).



NOTICE

Use Aero Duster ms-222n with the can held in the upright position only. Liquid discharged from the spray will result in static charge generation and possible damage to the detector. This would result in detector replacement. Frost or ice will be seen when this occurs.



NOTICE

The Aero Duster Spray System nozzle must be used, as it will prevent ESD damage to circuit board components. Hold the can in an upright position. Do not allow liquid to be sprayed onto any surface. The aerosol liquid evaporation can generate up to 12,000 volts of static charge.

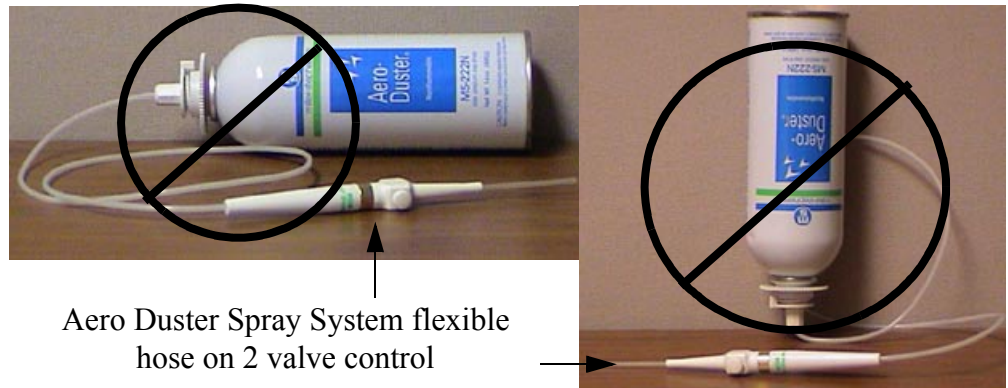


Figure 1-3 Incorrect Usage of AeroDuster

III.) Place in Anti-Static bag after inspection/cleaning.

Comment:

Start at slot 1 and proceed in order. Keep the original convertor cards in order so you can replace them in their original slots.



NOTICE

Do not vacuum the backplane connectors. This can result in damage to the detector from static discharge, resulting in a detector replacement.

- 5.) After all boards are removed, carefully vacuum the chassis interior and exterior if necessary. Pay particular attention to the MDAS Plenum interface.

4.3.2 Backplane Cleaning Procedure

- 1.) Connect ESD strap and ground to your person and ground point of rotating gantry. Ensure a good ground is present, no audio alert from ESD strap electronics.
- 2.) Carefully blow dust from the backplane of each of the three (3) chassis.

4.4 MDAS Assembly

- 1.) Connect ESD strap and ground to your person and ground point of rotating gantry. Ensure a good ground is present, no audio alert from ESD strap electronics.
- 2.) Install original convertor cards back in the original slots.
- 3.) Install new convertor cards in each of the remaining open slots.
- 4.) Using both hands ensure that each convertor card is fully seated. Approximately 1 mm after red ejector handles are closed.
- 5.) Install Chassis covers.
- 6.) Install DRY air filter into MDAS plenum.

4.5 Plenum Installation

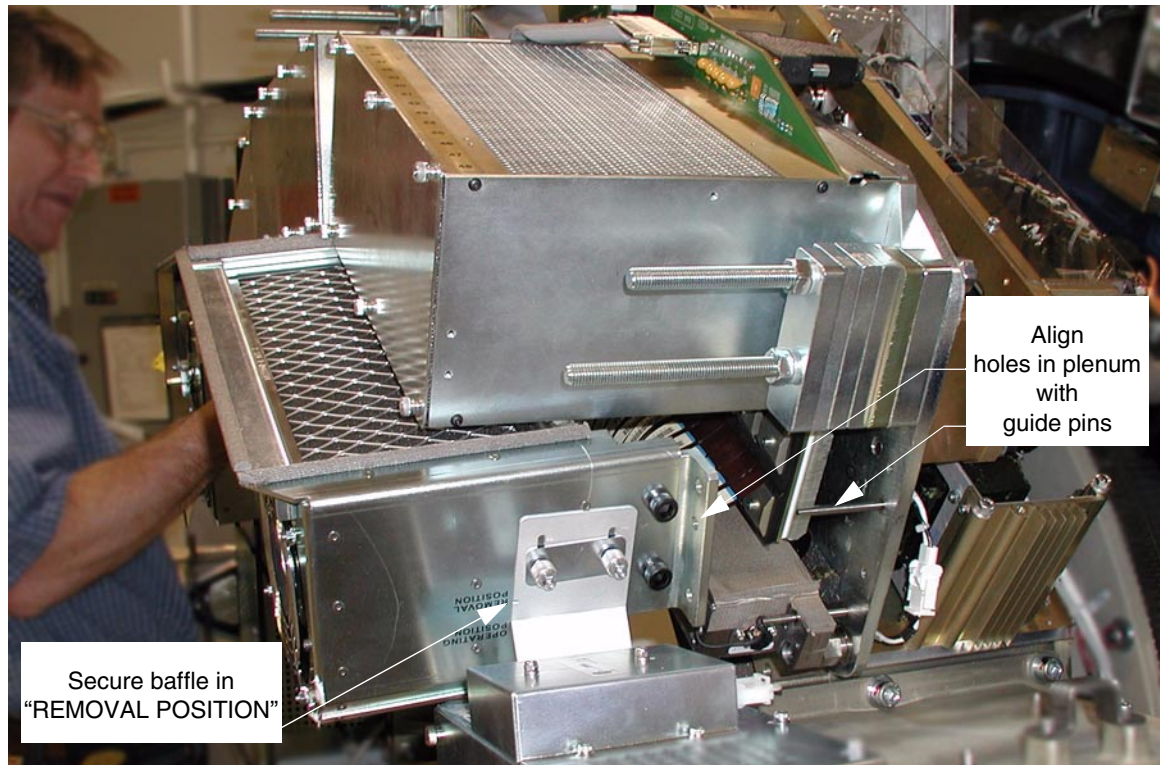


Figure 1-4 DAS Air Plenum Installation

- 1.) Ensure that baffles are secured in "REMOVAL POSITION".
- 2.) Lift plenum into place, and align guide holes with guide pins.
- 3.) Slide plenum onto guide pins.
- 4.) Fasten plenum securely into place, using the four (4) M6 mounting bolts and four lock washers. Torque to 7.9 N-m (69.9 in-lbs).
- 5.) Return baffles to the "OPERATING POSITION".
- 6.) Connect Fan AC power at Connector J22.
- 7.) Restore gantry 120 vac power at STC chassis.
Ensure MDAS fans turn ON and MDAS power up test occurs via LEDs on right chassis backplane.
- 8.) Proceed to DAS Power Supply Check.

4.6 DAS Power Supply Check

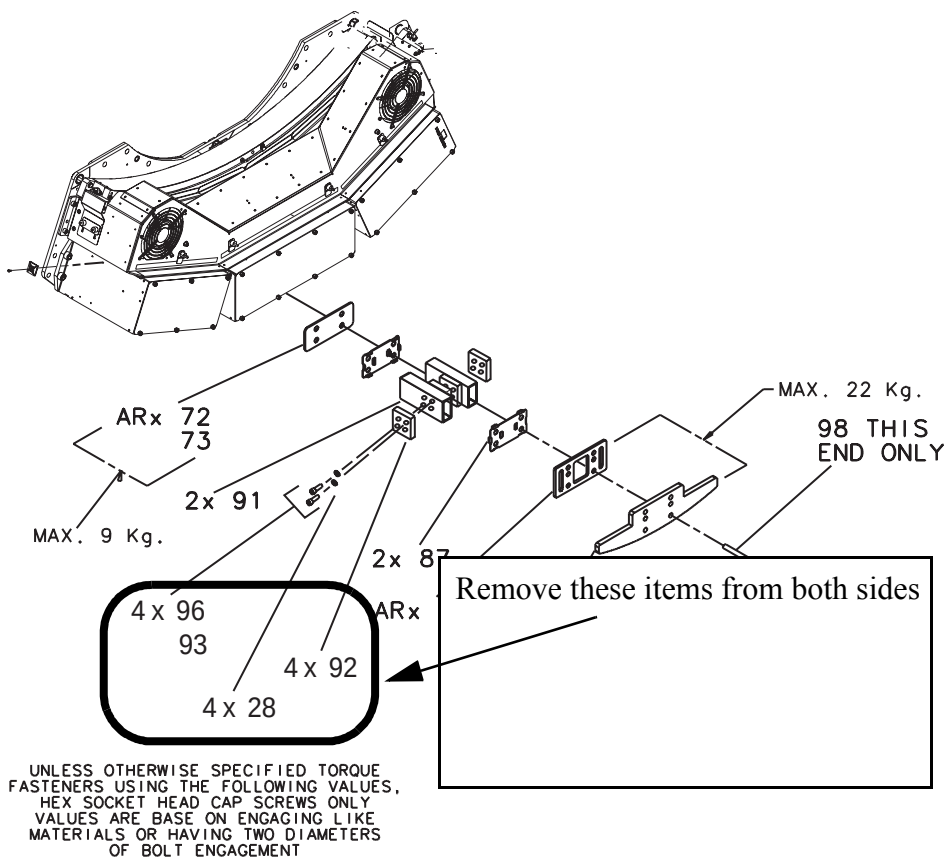
Since the load has in effect been doubled, you need to verify and adjust, as necessary, the DAS Power Supplies.

- Refer to the *LTSPD Ultra & Plus Service Manual - General* Direction 2296434-100, REVISION 4, Chapter 8 Gantry, Section 2.1.4, for specifications.
- Refer to the *LTSPD Ultra & Plus Service Manual - General* Direction 2296434-100, REVISION 4, Chapter 7 Detector and DAS, Section 3.4 for adjustment procedure details.
- Proceed to Balance Weight Removal Procedure.
This will allow time for the MDAS and Detector to warm up. It takes two (2) minutes of warm-up time for every minute the **DAS** was turned OFF, up to one (1) hour.

4.7 Balance Weight Removal Procedure

- 1.) Remove the four (4) balance weights as shown in [Figure 1-5](#).
 - Item 92, 96, and 28.
 - The two (2) M6 screws pass through the outer weight, the tube, and screw into the inner weight.
 - Be careful not to drop either the weights or the fastening hardware.

2291276-2 HELIOS 3 ROTATING BASE (4 SLICE) ASM ONLY



FASTENER SIZE	TORQUE
M3	1 Nm
M4	2, 3 Nm
M6	7, 9 Nm
M10	38, 4 Nm
M12	66, 4 Nm
M16	160 Nm

- Item 96 Screw Hexagon Socket 4MM
- Item 93 Loctite 242
- Item 92 Weight, MDAS 4 Slice Only Balance
- Item 28 Washer, Lock Spring 4.1 MM 7.6MM Spring Steel
- Item 54 Nut Hexagon 12 MM
- Item 47 Washer, Lock Spring 12.2 MM 21.1 MM

Figure 1-5 4 Slice Balance Weight Location

- 2.) Disengage the Rotational Lock.
- 3.) At the STC turn on HVDC and Axial Enable.
- 4.) Perform Service Gantry Balance found in Replacement Procedures of the Common Service Desktop.

If balance check fails, follow the instructions within the Service Gantry Balance program.

Chapter 2

Software Configuration

Purpose: In this section you will need to reconfigure the software, enable the 8 slice option with an options key, functional testing, and phantom calibrations.

REFER TO THE LIGHTSPEED ULTRA/PLUS SOFTWARE INSTALLATION PROCEDURES, DIRECTION 2296436-100, REVISION 02 FOR DETAILED RECONFIG INSTRUCTIONS.

REFER TO THE LIGHTSPEED 3.X APPLICATION TIPS AND WORK-AROUNDS, DIRECTION 2296447-100, REV 3, PAGE 12 FOR DETAILS ON UPDATING THE SCANNING PROTOCOLS.

REFER TO LTSPD ULTRA/PLUS SERVICE MANUAL - GENERAL, DIRECTION 2296434-100, REVISION 4, CHAPTER 11, IMAGE QUALITY FOR DETAILS. CHAPTER 2, SECTION 3.2.2 FLASH DOWNLOAD TOOL FOR DETAILS. CHAPTER 2, SECTION 2.0 SCANNER UTILITIES FOR CALIBRATION DETAILS.

- 1.) Shutdown Applications software.
- 2.) Perform Reconfig procedure.
- 3.) Select **MDAS8** from the Hardware Configuration Screen.
- 4.) Bring system to Applications Level.
- 5.) Perform Flash Download to re-initialize the DAS configuration.
- 6.) Install the 8 Slice Option, using the provided MOD, as instructed in the Software Load manual.
- 7.) Open **DASTOOLS** from the Service Desktop.
 - a.) Perform 1 pass **INTERCONNECT/XRAY VER TEST.**
 - b.) Perform 1 pass **POP NOISE & MICROPHONICS TEST.**
 - c.) Correct any failures.

Refer to the Appendix for De-Ionizing Procedure Details.
- 8.) Perform **DASGAIN CAL.**
- 9.) Perform **COLLIMATOR CAL.**
- 10.) Perform **FASTCAL.**
- 11.) Perform Detailed Phantom Calibrations.
- 12.) Perform 1x Image Series and evaluate Image Quality.

Chapter 3

Rating Plate Installation

There are 2 sets of rating plates that must be installed, Gantry, DAS Chassis and Console.

Section 1.0

Gantry Rating Plate

Install the UPGRADE TO 2281177 (8 SLICE) rating plate (2330483) on the left side of the gantry, next to the original manufactured rating plate, on the tilt support riser.

Section 2.0

DAS Chassis Rating Plate

Install the three (3) KIT, 4 TO 8 SLICE MDAS UPGRADE rating plate (2330482), one on each chassis next to the original manufactured rating plates. Do not cover any ventilation holes.

Section 3.0

Patient 8 Slice Option Rating Plate

Install the SERIAL #, MODEL#: rating plate (2316996) on the console next to the original manufactured rating plate.

Appendix

CT MDAS Convertor Card Cleaning & De-Ionizing Procedure

Purpose: The following procedure should be followed when **DAS** Converter boards and/or chassis need cleaning, or Converter boards are replaced for microphonics or noise resulting in image artifacts (rings, bands, streaks and smudges). This procedure is modified from the System Service Manual specifically for this upgrade procedure. For full procedure details reference the System Service manual.



WARNING

ROTATING GANTRY CAN CAUSE SEVERE INJURY OR DEATH. MAKE SURE GANTRY IS PROPERLY DE-ENERGIZED AND LOCKED WHILE PERFORMING THIS SERVICE. FOLLOW THE GANTRY SAFETY INSTRUCTIONS IN CHAPTER 1 OF THE SYSTEM SERVICE MANUAL.



NOTICE

Mishandling can easily damage converter cards. Handle only on the sides and only with proper ESD protection. Do not touch the connector, any board components, or circuit traces. Fingerprints are resistive circuit elements at low detector signal levels. Cards should always be placed in a Static Bag when not in the DAS.



CAUTION Chemicals

Store Amax Cleaner at Site with MSDS document. Do not store in the company car. Use spray as directed in this procedure. Safety glasses and finger cots must be worn when using the spray.



NOTICE

Do not clean the elastomers with AMAX spray. The elastomers absorb liquid. Their manufacturer does not recommend trying to clean elastomers.



NOTICE

When using compressed air, do not invert the can as this will cause liquid to come out. Testing has shown that this can cause a large static charge to develop where the spray hits.



NOTICE

Failure to use the Aero Duster spray system and proper ESD grounding will generate a static charge of DAS components up to 12,000 volts.



PROCEDURE

- 1.) Ensure system preparation is followed.
- 2.) All protective **ESD** materials should be in place (i.e. wrist straps and grounded mats for laying out converter cards).
- 3.) Remove the Converter cards (see [Figure A-1](#)) one at a time from their static bags and spray the connector (see [Figure A-2](#)) with it facing down. Spray the outside of the connector shroud on all sides and also spray inside the pin housing. (Give it a good soaking). Allow the card to air dry. Do not use the compressed air to dry it.
- 4.) To clean the card components, hold the card sideways (see [Figure A-1](#)) and spray from top to bottom. Direct spray away from edge connector. Allow card to air dry. Do not use compressed to dry it.

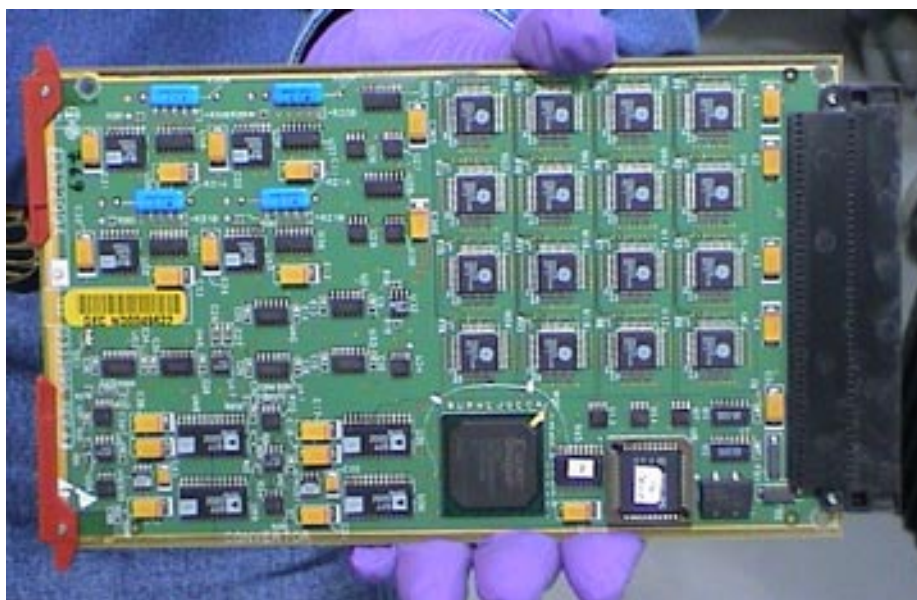


Figure A-1 Proper method for handling converter cards

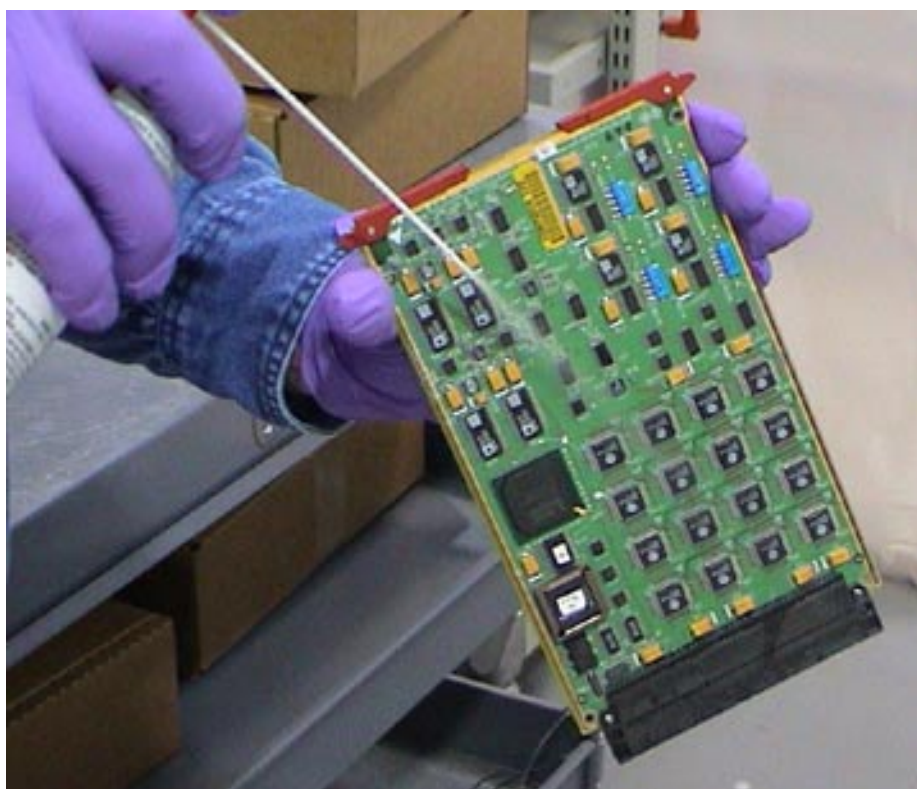


Figure A-2 Spraying "off" the connectors

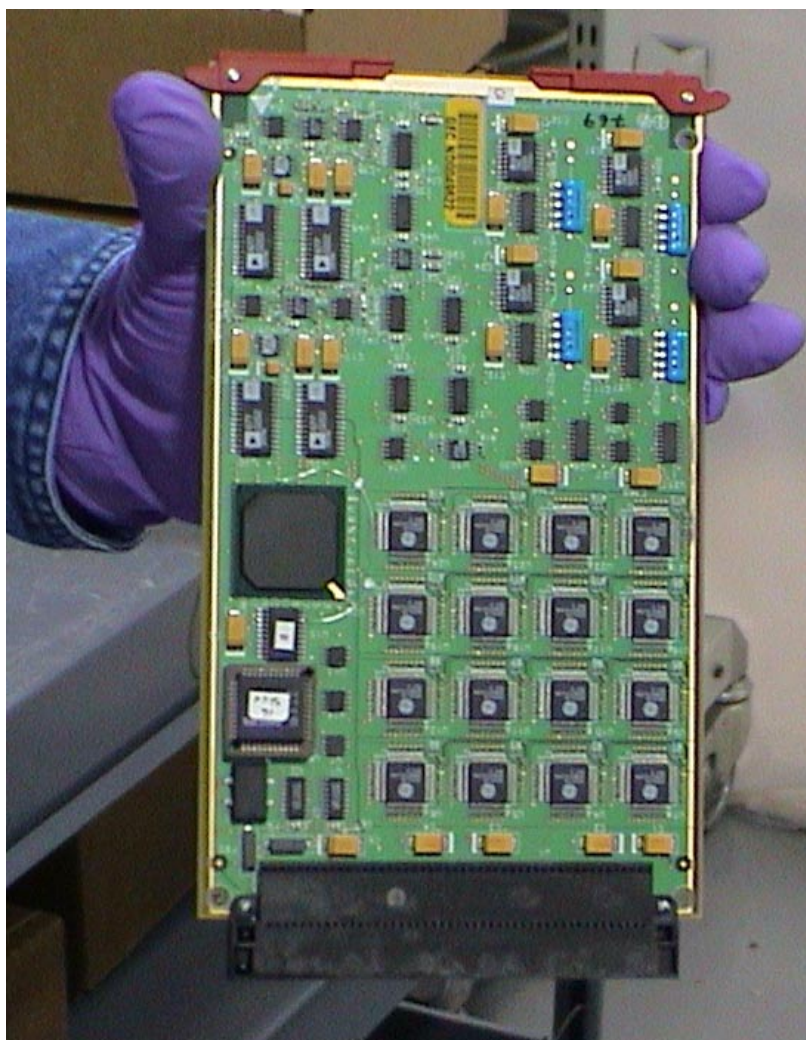


Figure A-3 Allow the cleaner to run off the converter card completely

- 5.) Reinstall cards in the same slots from which they were taken.

Turn the **DAS** ON and let it warm up. It takes two (2) minutes of warm-up time for every minute the **DAS** was turned OFF, up to one (1) hour.

Verify all covers are installed, turn on **DAS** power and complete the following tests within DASTools.

- 1 iteration of DC Means/Noise
- 1 iteration of microphonics/pop

All tests must pass. If not, troubleshoot and correct the failures.



GE MEDICAL SYSTEMS

***GE MEDICAL SYSTEMS-AMERICAS: FAX 262.312.7434
3000 N. GRANDVIEW BLVD., WAUKESHA, WI 53188 U.S.A.***

***GE MEDICAL SYSTEMS-EUROPE: FAX 33.1.40.93.33.33
PARIS, FRANCE***

***GE MEDICAL SYSTEMS-ASIA: FAX 65.291.7006
SINGAPOR***